

Program:	B. Tech. (Computer Engineering)						Semester: VI				
Course:	Design and Analysis of Algorithm						Code: BCE26PC02				
Credit	Teaching Scheme (Hrs./Week)				Evaluation Scheme and Marks						
	Lecture	Practical	Tutorial	Other	FA		SA	TW	OR	PR	Total
					FA1	FA2					
3	2	2	-	1	10	10	30	50	-	-	100

Prior knowledge of Data Structure is essential

Course Objectives:

This course aims at enabling students:

1. To develop problem solving abilities using mathematical theories
2. To analyze the performance of algorithms
3. To apply algorithmic strategies while solving problems
4. To develop time and space efficient algorithms
5. Apply Strategy-Specific Algorithms to Real-World Problems

Course Outcomes:

After learning the course, the students will be able to:

1. Analyze the efficiency of algorithms using mathematical techniques to evaluate algorithm performance.
2. Analyze the efficiency of algorithm using either Divide and Conquer or Greedy strategy
3. Apply computational techniques to solve optimization and decision-making problems using either Dynamic programming or Branch and Bound.
4. Apply Backtracking approach to implement solutions for combinatorial problems.

Guidelines for Laboratory/Term Work Assessment:

The laboratory assignments are to be submitted by students in the form of a journal.

1. Each assignment write-up should have Title, Objectives, Outcomes, Theory- Concept in brief, data description, conclusion, and assessor's sign.
2. Program codes with sample output of all performed assignments should be submitted.
3. Continuous assessment of laboratory work is done based on the overall performance and laboratory performance of the students.
4. Each laboratory assignment assessment should assign grade/marks based on parameters with appropriate weightage.
5. Suggested parameters for overall assessment as well as each laboratory assignment assessment include timely completion, performance, innovation, efficiency, punctuality and neatness.
6. All assignments are mandatory.
7. Recommended Tools for the implementation of above assignments: Python, Java etc.

Best Practices: GitHub, Leetcode, Hacker Rank and other similar platforms.

Detailed Syllabus

Unit	Description	Duration (Hrs)
I	Problem solving and Algorithmic Analysis: Asymptotic notations, lower bound and upper bound: Best case, worst case, average case analysis, classification of time complexities (linear, logarithmic etc)-Growth of function, Recurrences: Formulation and solving recurrence equations using Master Theorem, Introduction: P,NP, NP hard, NP Complete.	8

II	Divide and Conquer Strategy: Problem subdivision – Divide and Conquer: Binary search, Quick sort, Integer Arithmetic- Multiplication. Greedy strategy: Principle, control abstraction, time analysis of control abstraction, knapsack problem, Job scheduling algorithm with deadlines, Single Source Shortest Path Dijkstra's.	7
III	Dynamic Programming Strategy: Principle, control abstraction, time analysis of control abstraction, binomial coefficients, multistage graphs, 0/1 knapsack, All Pair Shortest Path - Floyd- Warshall, Bellman-Ford, Sum of subset.	8
IV	Backtracking Strategy: Principle, control abstraction, 8-queen problem, graph coloring problem, knapsack problem Branch and Bound Strategy: Principle, control abstraction, strategies – FIFO, LIFO and LC approaches, knapsack problem(FIFO,LIFO,LC), Branch and Bound using TSP	7
	Self Study: Solving recurrence relation using substitution and Tree method. Implement Selection sort, Insertion sort, Merge Sort, Quick Sort and compare performance. Solution and Comparative analysis of one example with all strategies e.g. knapsack problem, sum of subset problem etc., research paper reading.	6*
Total (*Not Included)		30
Suggested Lab Assignments:		
1. You are working as a data analyst for a financial institution that handles high-precision transactions involving large, multi-digit monetary values. Recently, the institution needs to calculate the product of two exceptionally large numbers (each over 15 digits) representing aggregated transaction amounts for quarterly reports. Due to the size of these values, traditional multiplication methods are computationally inefficient. Implement the algorithm to perform high-precision multiplication of these large integers accurately and efficiently. Test your implementation with multiple large-number pairs to ensure reliability and precision in financial reporting. 2. Design a greedy algorithm to maximize profit in a warehouse by selecting and transporting items to a single destination within a given weight limit. Each item has a specified weight and profit. Implement the algorithm and verify that it maximizes profit while staying within the weight constraint. 3. You find yourself in a labyrinthine dungeon with multiple rooms and connecting passageways. Each passage has a specific time to travel, and some passages are blocked due to cave-ins. Starting from the main entrance, you must find the shortest escape routes to every other room in the dungeon. Use Dijkstra's algorithm to determine the quickest time required to reach each room from the entrance. If any room is isolated and unreachable, mark it as -1. Test this with a dungeon of at least 15 rooms and random travel times for each accessible passage. 4. You are on a treasure hunt and have a map that shows various caves connected by tunnels. Each tunnel has a different cost to cross, representing difficulty or danger levels. You need to calculate the safest path between each pair of caves (i.e., the path with the minimum danger). Use the Floyd-Warshall		

